

Mridula Mahendran

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Education

Northeastern University

Master of Science, Information Systems

Anna University

Bachelor of Technology, Information Technology

Boston, MA

Aug 2026

Chennai, India

May 2023

Work Experience

Siemens

Software Developer Co-op

Plano, TX

June 2025 – Dec 2025

- Developed reusable HMI/SCADA components for an industrial automation platform using Angular, TypeScript, and SIMPL, resolving 30+ UI defects during QA cycles and ensuring consistent rendering across multi-resolution displays
- Implemented REST API endpoints using Node.js with Express and Prisma ORM, adding input validation (Zod), pagination, and role-based access checks across 15+ routes for an HMI asset configuration management service
- Built the document ingestion pipeline for a RAG-based documentation assistant using Python, LangChain, pdfplumber, and Camelot, processing 120+ technical documents with text and table extraction into ChromaDB embeddings
- Containerized the RAG pipeline and FastAPI backend using Docker, deployed to AWS ECS Fargate with S3 for document storage and CloudWatch for logging, reducing manual deployment steps from 12 to a single command
- Established GitLab CI/CD pipelines on code.siemens.com for the team's Python services, automating linting, unit tests (pytest), Docker image builds, and deployments to dev/staging environments, cutting build-to-deploy time from 15 minutes to under 5 minutes

Accenture

Advanced App Engineering Associate

Chennai, India

Nov 2023 – Aug 2024

Contributed to full-stack development and production support of a retail banking platform as part of a 12-member cross-functional Agile team.

- Engineered customer-facing features for the online banking portal using React 18, TypeScript, and Redux Toolkit, including transaction history redesign with advanced filtering and CSV export for 100K+ monthly active users
- Implemented API enhancements for the payments module using Spring Boot 3 and Spring Data JPA, adding input validation and standardized error handling that reduced invalid request errors in QA by 30%
- Participated in migrating three legacy SOAP-based services to REST APIs on Spring Boot, mapping existing WSDL contracts to OpenAPI 3.0 specifications and coordinating with downstream consumer teams during the cutover
- Monitored application health across 8 microservices using AppDynamics and Grafana, configuring 12+ alerts based on team-defined SLOs that helped detect a connection pool exhaustion issue before customer impact
- Contributed to bi-weekly client demos and sprint reviews, documented API contracts in Confluence, and supported knowledge transfer sessions during a team expansion from 12 to 18 engineers

Projects

Cloud Native Web Application | *AWS, Terraform, Packer, Shell Scripting, Node.js, Express.js, REST APIs, PostgreSQL*

- Architected and delivered a cloud-native web application using Node.js, Express.js, Sequelize, and PostgreSQL, deployed on AWS with S3 for file storage, optimized for scalability and high availability
- Provisioned end-to-end AWS infrastructure using Terraform, including VPCs, EC2 instances, security groups, Application Load Balancer, Auto Scaling Groups, and Route 53 DNS for fault tolerance and elastic scaling
- Automated a CI/CD pipeline using GitHub Actions, enabling continuous testing, custom AMI creation with Packer, and seamless deployments, reducing manual release effort and improving deployment reliability
- Configured secure AWS networking with Terraform, managing SSL certificates via ACM, IAM roles, firewall rules, and private RDS connectivity to ensure data privacy across dev and production environments

DataFyre | *MongoDB, Express.js, React.js, Node.js, JWT, ApexCharts.js, i18n, Redux, ANT Design Framework*

- Built a scalable synthetic data generation platform using the MERN stack, supporting API-driven data customization, live dashboards, and analytics tracking with ApexCharts.js and Redux for state management
- Integrated role-based access control, email-based authentication, and internationalization (i18n) using React, Node.js, and JWT, maintaining secure and user-friendly application functionality across multiple locales
- Created dynamic forms with ANT Design framework, instant mock data previews, and leveraged Bull for background task management, enhancing data processing workflows and reducing manual data generation effort

Employee Dashboard | *Java, Spring Boot, Hibernate, TypeScript, Angular, MySQL, Chart.js, REST APIs*

- Designed a centralized employee management system using Spring Boot, Hibernate, and Angular, providing secure storage and retrieval of employee data in a MySQL database through RESTful APIs
- Visualized department-specific statistics using Chart.js with real-time data fetched via REST API calls, driving informed decision-making through interactive charts and graphs

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, Shell Scripting

Frontend: React, Angular, Redux Toolkit, HTML/CSS/SCSS, ANT Design, Chart.js, ApexCharts.js

Backend: Node.js, Express.js, OpenAPI/Swagger, Spring Boot, FastAPI, Hibernate, Sequelize, Prisma ORM

Databases: PostgreSQL, MySQL, MongoDB, ChromaDB

Cloud & DevOps: AWS (EC2, S3, ECS Fargate, RDS, Route 53, CloudWatch, ALB, ACM, IAM, Auto Scaling), Terraform, Docker, Packer, GitHub Actions, GitLab CI/CD, Grafana

Core Competencies: Data Structures, Algorithms, REST APIs, OOP, Design Patterns, Microservices Architecture, Agile